UK Data Service

FAIRly Specialist?

Generic vs Disciplinary implications for FAIR-enabling Repositories

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IASSIST, 2025
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Next?

As is somewhat traditional, the answers we were hoping to provide when the abstract was submitted have been replaced by another series of questions.

Working Paper, comment, critique, add your ORCID

https://bit.ly/FAIRly

https://docs.google.com/document/d/1jgWRWXjtuu4BGWyG59H8xO_hTRMgQ8bndVXUZz7xm2c/edit?usp=sharing

Related: FIDELIS Transparent Trustworthy Repository Attributes Matrix (TTRAM). Workshop June 17th

http://bit.ly/3FFv3Pu

https://eden-fidelis.eu/events/introduction-ttram-matrix-webinar-0

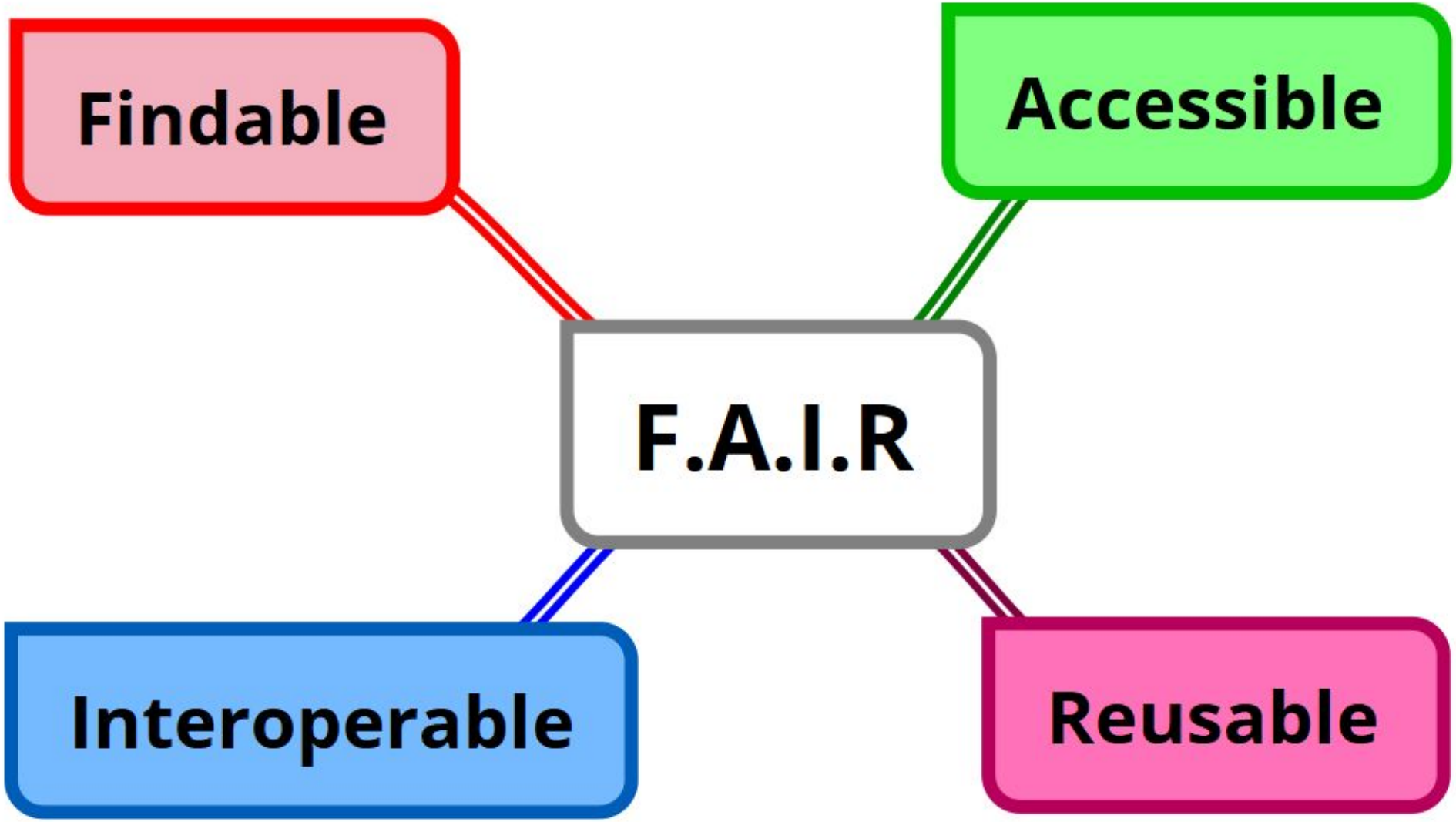
Hat & Perspectives

Social Sciences

- FSD, the Finnish Social Science Data Archive, Tampere University
- DANS, Data Archiving and Networked Services in the Netherlands
- UKDS Multi-Partner ESRC Social Science Data Service
- UKDA Lead Partner and Trustworthy Digital Repository
- CESSDA Consortium of Social Science Data Archives (ERIC)
  - CESSDA Trust & Landscape group
- ESRC Future Data Services

Beyond Social Sciences

- FAIR IMPACT project (now complete)
- EOSC EDEN project on preservation
- FIDELIS building a network of Trustworthy Repositories
- OSTrails project on FAIRness and Planning, Tracking and Assessing
- EOSC Association FAIR Metrics and Digital Objects Task Force
- EOSC Association Long Term Retention Task Force
  - Following on from the Long Term Preservation Task Force


Findable
Accessible
F.A.I.R
Interoperable
Reusable

FAIR Findable, Accessible, Interoperable, ReUsable

The FAIR Principles actually embody one key ‘Principle’:

> data and metadata (*all digital objects*) should be findable, accessible, interoperable and reusable.

There are 15 narrow criteria (the ‘Principles’) with a focus on machine actionability.

FAIR Principles

Findable

- F1. Meta(data) Identifier: Globally Unique/Persistent
- F2. Rich Metadata Descriptions
- F3. Metadata: Specifies Data Identifier
- F4. Searchable Resource: Indexed/Registered

Accessible

- A1. Meta(data):
  *Retrievable via ID
  *Standard Comms Protocols
  - A1.1. Protocol: Open, Free, Universal
  - A1.2. Protocol: Auth/Auth (if necessary)
- A2. Metadata Remain Accessible, when Data not Available

Interoperable

- I1. Meta(data) Knowledge Representation:
  *Formal
  *Accessible
  *Shared
  *Broadly Applicable Language
- I3. Meta(data) Qualified References to other (Meta)data

Reusable

- R1. Meta(data) Plurality of Attributes:
  *Rich
  *Accurate
  *Relevant
  - R1.1 Meta(data) Data Usage License:
    *Clear
    *Accessible
  - R1.2 Meta(data) Provenance
  - R1.2 Meta(data) Domain-relevant Community Standards
- I2. Meta(data) Vocabularies: Follow FAIR Principles

FAIR and Repositories

For repositories, **enabling** FAIR digital objects has been added to the range of outcomes expected from their retention, curation and preservation of digital assets relevant to research.

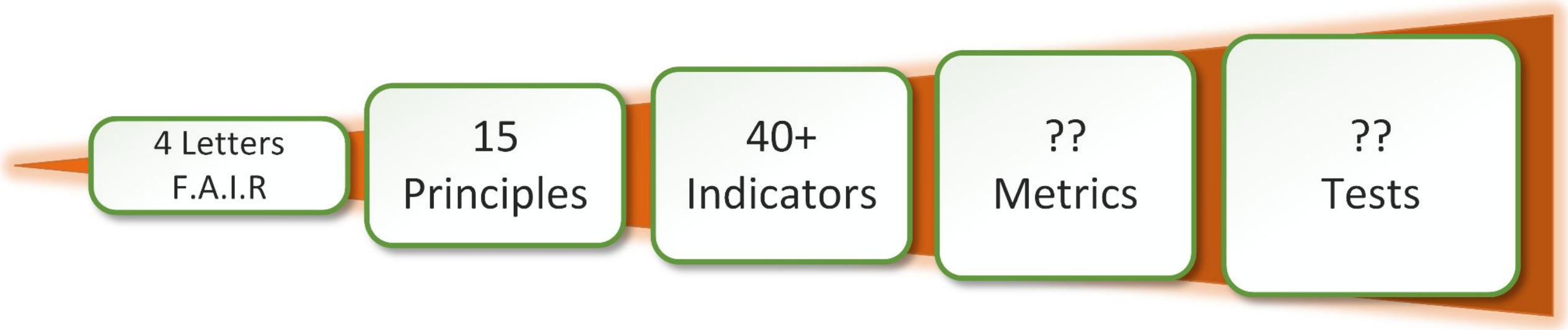
- Risk of narrowing focus (and funding) from wider aspects:
  - Finding
  - Accessing
  - Interoperating with
  - Reusing

FAIR Objects, Indicators, Metrics and Tests

The Principles are supported by ‘indicators’, including those developed through the Research Data Alliance.

A wide range of metrics have been developed against these, and other interpretations of the Principles.

These have been used in a range of tools that test FAIRness at object level.



HLH-FAIR-AcronymPrinciplesIndicatorsMetricsTests_02_00.vsdx

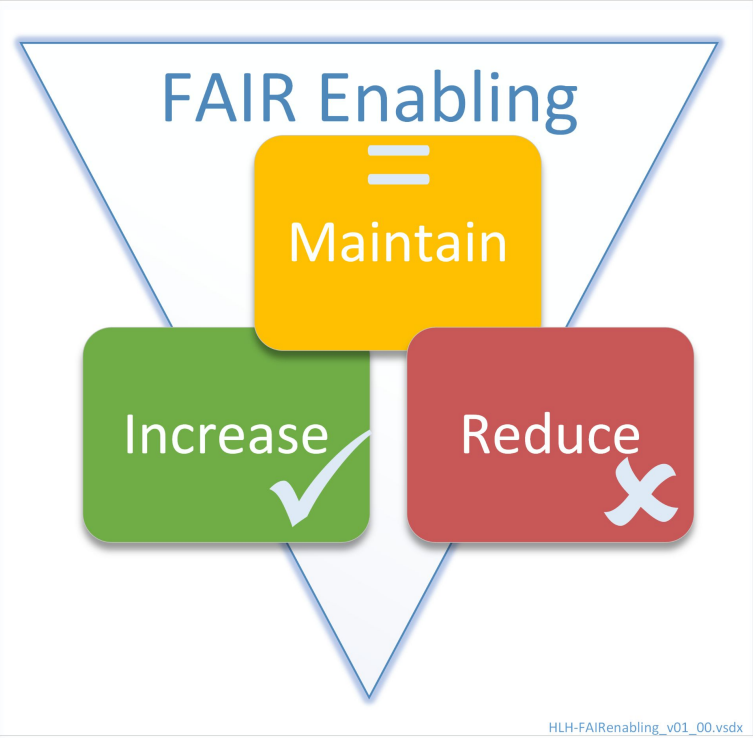
Tests, Objects and Repositories

Tests are applied to individual digital objects including traditional research data, ontologies and software.

But it's the repositories that provide the environment to enable FAIRness

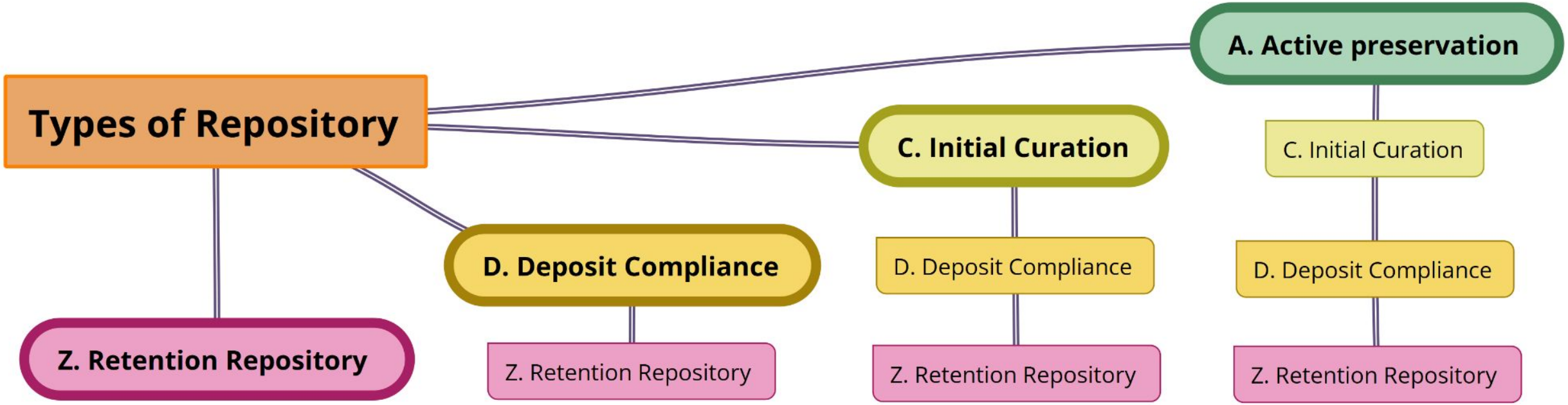
Repositories must:

- **Maintain** FAIRness from the point of deposit
- **Increase** FAIRness during curation and preservation
- Avoid changes that **reduce** FAIRness



FAIR + Time & Levels of Care

FAIR is a snapshot, the principles are not time-sensitive

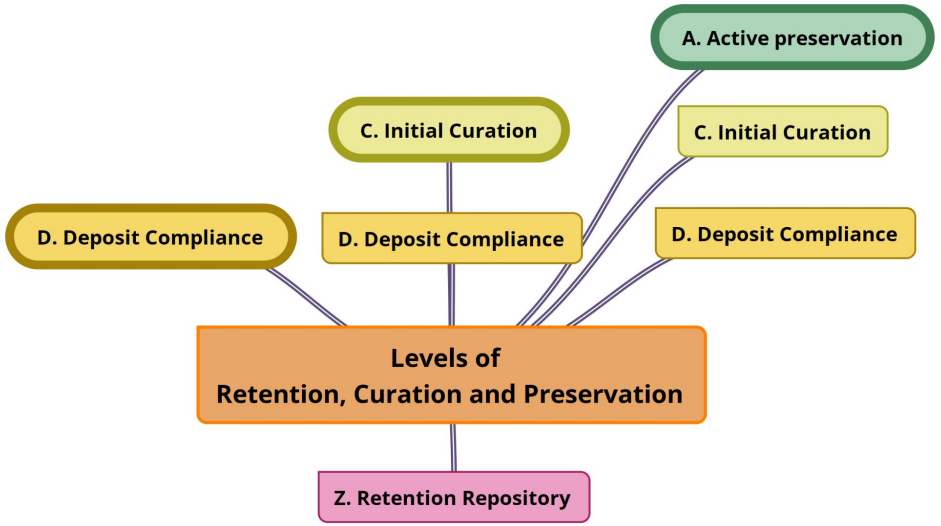


Intervention points and time-based guarantees

Beyond basic retention of digital objects for a defined period of time, different repositories provide different levels of care to digital objects:

- Checking **compliance** at the point of **deposit**
- Ensuring compliance through **initial curation** actions
- Undertaking to maintain compliance over time through **active preservation**

Compliance could be with any defined criteria, including FAIR.



FAIR + Time & Trustworthy Repositories

F R12

F1. (meta)data are assigned a globally unique and persistent identifier.
F2. data are described with rich metadata (defined by R1 below).
F3. metadata clearly and explicitly include the identifier of the data it describes.
F4. (meta)data are registered or indexed in a searchable resource.
Discovery & Identification (R12)

A R09 R15 R16

A1 (meta)data are retrievable by their identifier using a standardized communications protocol.
A1.1 the protocol is open, free, and universally implementable (vs context)
Technical Infrastructure (R15)
A1.2 the protocol allows for an authentication and authorization procedure, where necessary.
Security (R16)
A2 metadata are accessible, even when the data are no longer available.
Preservation Plan (R09)

I R13

I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
I2. (meta)data use vocabularies that follow FAIR principles.
I3. (meta)data include qualified references to other (meta)data.
Reuse (R13)

R R02 R07 R14

R1. meta(data) are richly described with a plurality of accurate and relevant attributes.
Reuse (R13)
R1.1. (meta)data are released with a clear and accessible data usage license.
Rights Management (R02)
R1.2. (meta)data are associated with detailed provenance.
Provenance and Authenticity (R07)
R1.3. (meta)data meet domain-relevant community standards.
Reuse (R13)

HLH-CoreTrustSeal-2023-25-FAIRenabling-Map_v04_01.vsdx

The CoreTrustSeal Trustworthy Digital Repository certification assesses active preservation repositories against 16 criteria:

- digital object management
- organisational infrastructure
- technology and security

Mapped to FAIR-enabling.

Disciplinary FAIR

- European Open Science Cloud (EOSC) federating research infrastructures
- Focus on FAIR alignment with disciplinary (Social Science) needs
- FAIR IMPACT project and F-UJI FAIR assessment tools developers worked with CESSDA
- Produced a version of the tool that integrated both generic and disciplinary level tests.

FAIR, Tests & Dependencies

Principles range from the more **specific** and **testable**

- F1 ‘assign an identifier’

…to the more **general** and **aspirational**

- R1 ‘*richly* described with a *plurality* of accurate and *relevant* attributes’.

…to some more explicit about the need for **external references**

- R1.3. (meta)data meet *domain-relevant community standards*

F-UJI, FAIR and CESSDA Social Science Repositories

Generic versus Disciplinary by Principle

No difference for many:

- F1. (Meta)data are assigned a globally unique and persistent identifier

This is a broadly applicable technical test that PIDs align with **expected** providers

Changes from generic to disciplinary in others:

- F2. Data are described with rich metadata

Supports a narrower and more specific test of which of ‘core metadata’ fields are **expected**.

- F4. (Meta)data are registered or indexed in a searchable resource

Disciplinary tests look for a narrower set of **expected** schemas and serialisations for harvesting.

F-UJI, FAIR and CESSDA Social Science Repositories

Generic versus Disciplinary by Principle

- I2. (Meta)data use vocabularies that follow FAIR principles
- R1.3. (Meta)data meet domain-relevant community standards

Tests for **expected** discipline specific vocabularies and data file formats.

Expected…expected…expected…

F-UJI and other FAIR Tests

Applying the tool to sample datasets:

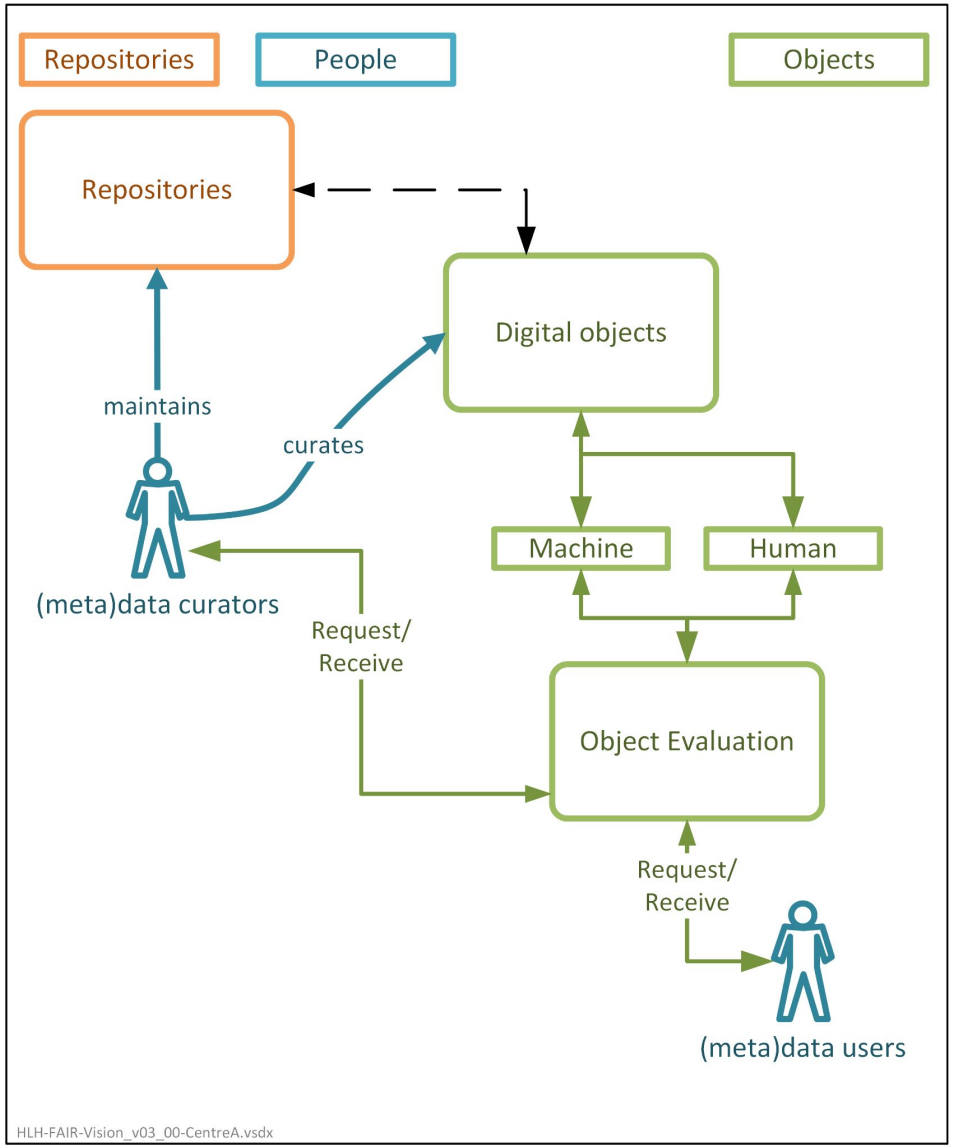
- Sometimes a generic test is repeated for social science
- Sometimes an equivalent test is absent.
- Differences in total score models between generic and disciplinary meant social science could score lower.
  - Repositories could just choose a test (or tool) that scores them higher.
- No standard approach across the 20+ FAIR tools, though efforts are being made, e.g OSTrails Project
- Single tests for single objects, no batch processing or scaling
- Imply repository use at deposit, and after curation and preservation actions.

Repositories and FAIR-enabling

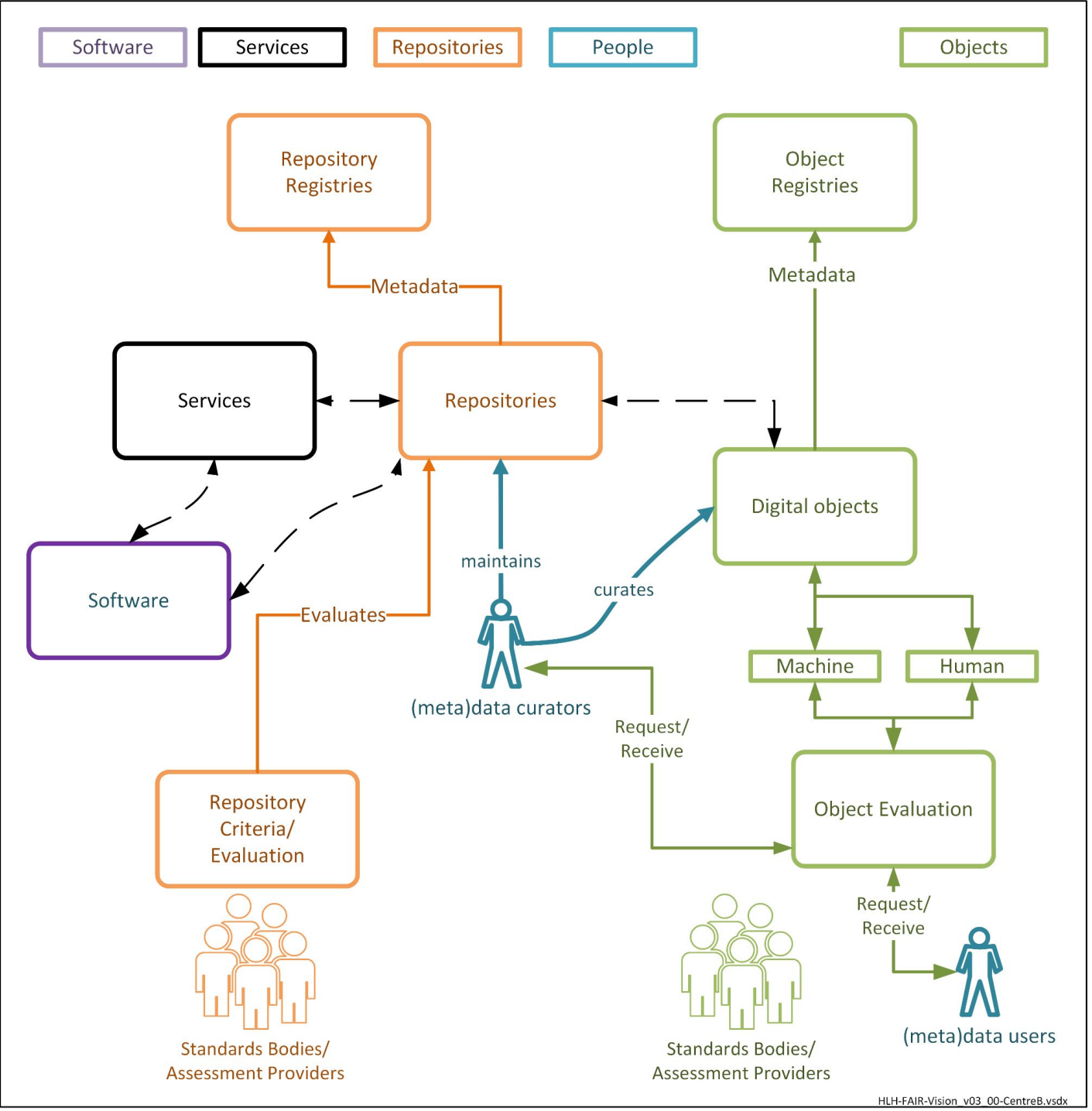
Challenges for repositories seeking to be FAIR-enabling:

- Tests don't address who has the authority to say a particular result is **expected**.
  - CESSDA not a global authority on SSH formats, metadata and ontologies.
  - The tests don't support a clear authority model.
- A repository cannot declare "all objects have PIDs" because tests are at object level
  - No method to test repository level metadata "Yes we PID all our objects"
- There is no standard guidance on how to expose object metadata for testing
  - Being 'interoperable' with a FAIR Assessment tool isn't a guarantee of general machine-actionability.

A Wider Perspective on Assessment



To enable FAIRness and assess outcomes we need to integrate repositories and their role in digital object assessment into a wider context.



This includes the software and services that support repositories, repository assessment criteria, and the registries that describe object and repositories.

Repositories, Assessment and Discipline

Can repositories deploy current FAIR-assessment tools into workflows?

- Need guidance on exposing repositories/object information to support
  - transparency in general
  - assessment in particular.
- FAIR is one of many standards, criteria, requirements, policies, guidance and best practices that repositories are expected to comply with.
  - All of these have different authorities, criteria, metrics and tests.

Any assessment needs to be considered alongside the level of curation and preservation provided.

To Disciplines and Beyond! [Domains are not the only fruit]

- IASSIST is a social science audience; the authors represent social science data archives.
- Repository checks on digital objects can include completeness of metadata, ontologies and use of appropriate file formats.
- These are often seen as key areas where disciplinary repositories will have more specific criteria.

But is FAIR Assessment by:

`'repositories where type = disciplinary'`

an oversimplification of real world scenarios?

Generic and Disciplinary Types

“Generic” repositories: important and necessary for the scale of research data.

- High-level, standardised, harvestable metadata supports interoperability
  - specialist repositories should too

For researchers and funders of disciplinary research the best outcomes for discovery, access, reuse and credit to authors comes from depositing in “disciplinary” repositories.

Generic and Disciplinary Types

FIDELIS project working towards a network of trustworthy repositories

- Stating a basic repository type (generic, disciplinary, institutional, national etc) doesn't tell us everything we need to know about them.
- No single coherent definition of a generic/generalist repository
  - but should they be described as 'multi-disciplinary'?
- No single definition of a disciplinary repository beyond self-labelling.

The EDEN project working on preservation

- Uses the Fields of Research and Development (FORD)
  - but many disciplinary controlled vocabularies exist.

The FAIR Principles actually refer to 'domain-relevant' **while the F-UJI metrics refer to a** 'target research community'**.**

FAIR IMPACT & EDEN Experience

FAIR IMPACT quotes:

"However, it very quickly became clear that these *use case communities were too broad*, as FAIR practices differed greatly and sub-communities could easily be identified, for example at *infrastructure and sub-discipline level*."

"*Preferences for FAIR metrics varied*, with some favouring large *discipline-specific* metrics, others focusing on *data types* or *research methods*, and a smaller group supporting *detailed metrics for sub-disciplines*. "

EOSC EDEN project desk research

Disciplinary repositories don't document that a specific approach, process or choice of format, metadata or ontology is for '*discipline-specific reasons'.*

Disciplinary-relevant versus Discipline-Specific

The Social Science & Humanities Open Cloud (SSHOC) identified common CESSDA Archive practice:

- Data in SPSS and STATA formats supported by
- Metadata in DDI (Data Documentation Initiative) using
- Vocabularies and Ontologies such as the European Languages Social Science Thesaurus (ELSST)

None are strictly limited to Social Sciences.

- FAIR only states: “domain-relevant community standards”
- Multiple communities and community standards within a domain.
- Multiple disciplines within a domain
- Multiple sub-disciplines

Then multi, cross, inter or trans disciplinary research, and what that actually means for the expectations from repositories.

Disciplinary-relevant versus Discipline-Specific

- Qualitative and quantitative research methods represent distinct expertise within Social Science but not unique to Social Sciences.
- Interviews, surveys and statistical analysis methods have some very specific tools and formats that aren't unique to Social Science.
- Geography-related formats, metadata and ontologies can be applied across a huge range of scientific research.
- Social Science researchers frequently record interviews, but are not A/V specialists.

Complex and challenging, best practices do exist,
but discipline is not the only perspective.

Multiple specialisms/forte across research methods and content types.

Types, Categories and Variables

Specialist: disciplines, methods, content types, file formats, metadata schemas and ontologies

Levels of Curation and Preservation: expected by funders, depositors and users, delivered by skilled repositories

- WHO can deposit/access
- WHICH objects in
- WHAT repositories

Depends on understanding how specialist metadata at the *object, depositor, user and repository* levels interact.

Conclusions, Communication and Choices

We need a more sophisticated approach to the different specialisms and forte that influence repository activities and functions, including discipline and domain, but also **methods, object types, file formats/serialisations, schemas and ontologies**.

- Pressure on repositories seeking to comply with multiple criteria from multiple **authorities**.
- **No single binary test** outcome defines a repository as FAIR-enabling
  - or trustworthy etc

We need

- **models that integrate contexts and expectations**
- **cooperative approaches and common registries**

Next?

Working Paper, comment, critique, add your ORCID

https://bit.ly/FAIRly

https://docs.google.com/document/d/1jgWRWXjtuu4BGWyG59H8xO_hTRMgQ8bndVXUZz7xm2c/edit?usp=sharing

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http://bit.ly/3FFv3Pu

https://eden-fidelis.eu/events/introduction-ttram-matrix-webinar-0

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